

DAS in Management of Clinical Trials

Module 2 · Principles and Methods of Clinical Research

Topic 7

# Descriptive statistics

Describing the data we observed

Mucyo Karemera · Stéphane Guerrier

Types of variables

Numerical summaries

Visualisation

# Statistics in a clinical trial

Hayward et al. · JAMA · 2017

Research

JAMA | Original Investigation

Effect of Oral Dexamethasone Without Immediate Antibiotics vs Placebo on Acute Sore Throat in Adults

A Randomized Clinical Trial

Gail Nicola Hayward, DPhil, MRCP, Alastair D. Hay, FRCP, Michael V. Moore, MSc, Sena Jawad, MSc, Nicola Williams, MSc, Meryn Voysey, MSc, Johanna Cook, BA, Julie Allen, BA, Matthew Thompson, DPhil, Paul Little, MD, Rafael Perera, DPhil, Jane Wooster-Holme, PhD, Kim Harman, DHealth, Carl Heneghan, DPhil

IMPORTANCE

Acute sore throat poses a significant burden on primary care and is a source of inappropriate antibiotic prescribing. Corticosteroids could be an alternative symptomatic treatment.

OBJECTIVE

To assess the clinical effectiveness of oral corticosteroids for acute sore throat in the absence of antibiotics.

DESIGN, SETTING, AND PARTICIPANTS

Double-blind, placebo-controlled randomized trial (April 2013–February 2015; 28-day follow-up completed April 2015) conducted in 42 family practices in South and West England, enrolled 576 adults recruited on the day of presentation to primary care with acute sore throat not requiring immediate antibiotic therapy.

INTERVENTIONS

Single oral dose of 10 mg of dexamethasone (n = 293) or identical placebo (n = 283).

MAIN OUTCOMES AND MEASURES

Primary: proportion of participants experiencing complete resolution of symptoms at 24 hours. Secondary: complete resolution at 48 hours, duration of moderately bad symptoms (based on a Likert scale, 0, normal; 5, as bad as it could be), visual analog symptom scales (0–100 mm; 0, no symptom to 100, worst imaginable), health care attendance, days missed from work or education, consumption of delayed antibiotics or other medications, adverse events.

RESULTS

Of 565 eligible randomized participants (median age, 34 years [interquartile range, 26.0–45.5 years]; 75.2% women; 100% completed the intervention), 288 received dexamethasone and 277, placebo. At 24 hours, participants receiving dexamethasone were not more likely than those receiving placebo to have complete symptom resolution. Results were similar among those who were not offered an antibiotic prescription and those who were offered a delayed antibiotic prescription. At 48 hours, more participants receiving dexamethasone than placebo had complete symptom resolution. This difference also was observed in patients not offered delayed antibiotics. There were no significant differences in any other secondary outcomes.

|                                                    | Total/No. (%)<br>Dexamethasone<br>(n = 288) | Placebo<br>(n = 277) | Risk Difference, %<br>(95% CI) | RR (95% CI)      | P Value |
|----------------------------------------------------|---------------------------------------------|----------------------|--------------------------------|------------------|---------|
| Resolution of Symptoms at 24 h (Primary Outcome)   |                                             |                      |                                |                  |         |
| Overall                                            | 65 (22.6)                                   | 49 (17.7)            | 4.7 (−1.8 to 11.2)             | 1.28 (0.82–1.78) | .14     |
| Resolution of Symptoms at 48 h (Secondary Outcome) |                                             |                      |                                |                  |         |
| Overall                                            | 102 (35.4)                                  | 75 (27.1)            | 8.7 (1.2 to 16.2)              | 1.31 (1.02–1.68) | .03     |
| No antibiotic prescription                         | 65/173 (37.6)                               | 46/169 (27.2)        | 10.3 (0.6 to 20.1)             | 1.37 (1.01–1.87) | .046    |

CONCLUSIONS AND RELEVANCE

Among adults presenting to primary care with acute sore throat, a single dose of oral dexamethasone compared with placebo did not increase the proportion of patients with resolution of symptoms at 24 hours. However, there was a significant difference at 48 hours.

TRIAL REGISTRATION

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JAMA 2017;317(15):1535–1543. doi:10.1001/jama.2017.3417

Author Video Interview

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**CONCLUSIONS AND RELEVANCE** Among adults presenting to primary care with acute sore throat, a single dose of oral dexamethasone compared with placebo did not increase the proportion of patients with resolution of symptoms at 24 hours. However, there was a significant difference at 48 hours.

How do the observed data support these clinical conclusions?

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1535

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# Descriptive statistics

Organise observations

A table may be enough for a small dataset.

With hundreds of participants:

graphs and numerical summaries.

JAMA Table 1 is already a summary.

Table 1. Baseline Characteristics of Eligible Patients

|                                          | All Patients               |                      |
|------------------------------------------|----------------------------|----------------------|
|                                          | Dexamethasone<br>(n = 288) | Placebo<br>(n = 277) |
| Age, median (IQR), y                     | 33.7 (26.3-45.8)           | 34.3 (26.0-45.0)     |
| Men, No. (%)                             | 67 (23.3)                  | 73 (26.4)            |
| Working or in school, No. (%)            | 217 (75.3)                 | 217 (78.3)           |
| Smoker, No. (%)                          | 52 (18.1)                  | 51 (18.4)            |
| Study center, No. (%)                    |                            |                      |
| Bristol                                  | 79 (27.4)                  | 72 (26.0)            |
| Oxford                                   | 143 (49.7)                 | 139 (50.2)           |
| Southampton                              | 66 (22.9)                  | 66 (23.8)            |
| Duration of sore throat,<br>mean (SD), d | 3.86 (1.67)                | 3.91 (1.79)          |

# Descriptive statistics

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| Oxford                                   | 143 (49.7)                 | 139 (50.2)           |
| Southampton                              | 66 (22.9)                  | 66 (23.8)            |
| Duration of sore throat,<br>mean (SD), d | 3.86 (1.67)                | 3.91 (1.79)          |

Describe the observed data before asking what we can conclude beyond them.

# Types of variables

Suppose we collect these data from *162* clinic visitors.

| ID  | Group | Severity | Visits | Wait | ID  | Group | Severity | Visits | Wait | ID  | Group | Severity | Visits | Wait | ID  | Group | Severity | Visits | Wait | ID  | Group | Severity | Visits | Wait | ID  | Group | Severity | Visits | Wait |
|-----|-------|----------|--------|------|-----|-------|----------|--------|------|-----|-------|----------|--------|------|-----|-------|----------|--------|------|-----|-------|----------|--------|------|-----|-------|----------|--------|------|
| 001 | A     | Severe   | 4      | 7.4  | 028 | O     | Severe   | 0      | 4.2  | 055 | O     | Moderate | 0      | 15.5 | 082 | B     | Severe   | 3      | 12.7 | 109 | O     | Moderate | 3      | 5.5  | 136 | A     | Mild     | 0      | 9.6  |
| 002 | O     | Mild     | 1      | 2.2  | 029 | A     | Mild     | 4      | 15.6 | 056 | A     | Severe   | 0      | 9.0  | 083 | O     | Mild     | 2      | 21.6 | 110 | A     | Mild     | 2      | 10.5 | 137 | O     | Severe   | 4      | 5.2  |
| 003 | B     | Mild     | 2      | 5.4  | 030 | A     | Severe   | 0      | 11.7 | 057 | O     | Mild     | 0      | 10.2 | 084 | O     | Severe   | 0      | 14.8 | 111 | O     | Severe   | 4      | 2.3  | 138 | A     | Moderate | 2      | 11.0 |
| 004 | A     | Moderate | 2      | 1.4  | 031 | O     | Moderate | 2      | 10.8 | 058 | O     | Severe   | 0      | 22.9 | 085 | O     | Mild     | 0      | 15.5 | 112 | A     | Severe   | 3      | 2.9  | 139 | O     | Mild     | 2      | 20.4 |
| 005 | AB    | Moderate | 2      | 4.6  | 032 | B     | Severe   | 2      | 18.4 | 059 | A     | Mild     | 4      | 4.9  | 086 | A     | Moderate | 4      | 17.6 | 113 | O     | Mild     | 4      | 6.8  | 140 | O     | Severe   | 2      | 16.7 |
| 006 | O     | Mild     | 3      | 6.2  | 033 | B     | Mild     | 2      | 5.5  | 060 | O     | Moderate | 2      | 11.5 | 087 | A     | Mild     | 2      | 2.8  | 114 | A     | Mild     | 3      | 21.3 | 141 | O     | Severe   | 3      | 4.6  |
| 007 | A     | Mild     | 0      | 3.8  | 034 | A     | Severe   | 0      | 10.8 | 061 | A     | Moderate | 0      | 18.2 | 088 | O     | Moderate | 0      | 2.6  | 115 | A     | Severe   | 1      | 14.6 | 142 | A     | Moderate | 1      | 15.2 |
| 008 | O     | Mild     | 1      | 21.0 | 035 | O     | Moderate | 3      | 21.4 | 062 | O     | Severe   | 1      | 7.7  | 089 | A     | Mild     | 4      | 17.1 | 116 | O     | Mild     | 4      | 10.3 | 143 | A     | Mild     | 4      | 13.8 |
| 009 | O     | Moderate | 1      | 16.5 | 036 | A     | Severe   | 4      | 1.6  | 063 | B     | Moderate | 3      | 9.3  | 090 | A     | Severe   | 4      | 11.7 | 117 | A     | Mild     | 0      | 10.7 | 144 | O     | Mild     | 3      | 6.8  |
| 010 | O     | Severe   | 1      | 2.0  | 037 | A     | Mild     | 2      | 11.2 | 064 | B     | Severe   | 2      | 22.8 | 091 | A     | Moderate | 0      | 20.7 | 118 | O     | Moderate | 0      | 5.6  | 145 | O     | Severe   | 4      | 13.6 |
| 011 | B     | Mild     | 0      | 21.2 | 038 | O     | Moderate | 0      | 21.7 | 065 | A     | Moderate | 3      | 17.1 | 092 | O     | Mild     | 3      | 17.2 | 119 | O     | Moderate | 3      | 2.7  | 146 | A     | Moderate | 1      | 6.9  |
| 012 | A     | Mild     | 0      | 4.4  | 039 | A     | Mild     | 2      | 2.5  | 066 | A     | Severe   | 1      | 11.7 | 093 | A     | Mild     | 0      | 17.4 | 120 | A     | Moderate | 1      | 14.7 | 147 | A     | Moderate | 1      | 19.0 |
| 013 | A     | Mild     | 3      | 6.7  | 040 | A     | Severe   | 4      | 22.5 | 067 | O     | Moderate | 4      | 11.1 | 094 | A     | Moderate | 1      | 19.7 | 121 | A     | Moderate | 4      | 21.5 | 148 | O     | Severe   | 4      | 11.2 |
| 014 | O     | Mild     | 4      | 4.0  | 041 | O     | Mild     | 2      | 1.8  | 068 | A     | Severe   | 2      | 16.1 | 095 | B     | Severe   | 1      | 10.7 | 122 | O     | Mild     | 0      | 20.5 | 149 | A     | Severe   | 4      | 16.2 |
| 015 | A     | Moderate | 3      | 18.7 | 042 | AB    | Severe   | 0      | 16.7 | 069 | A     | Mild     | 4      | 14.4 | 096 | O     | Moderate | 1      | 8.2  | 123 | A     | Severe   | 4      | 7.4  | 150 | O     | Mild     | 3      | 16.1 |
| 016 | O     | Moderate | 4      | 13.1 | 043 | B     | Severe   | 0      | 2.3  | 070 | A     | Severe   | 1      | 3.6  | 097 | O     | Mild     | 0      | 21.3 | 124 | A     | Mild     | 4      | 1.2  | 151 | O     | Severe   | 2      | 17.2 |
| 017 | O     | Mild     | 3      | 3.9  | 044 | AB    | Mild     | 0      | 19.8 | 071 | O     | Moderate | 4      | 16.1 | 098 | O     | Mild     | 0      | 3.0  | 125 | A     | Mild     | 1      | 8.7  | 152 | A     | Mild     | 3      | 13.2 |
| 018 | O     | Moderate | 3      | 21.1 | 045 | AB    | Moderate | 3      | 18.5 | 072 | A     | Severe   | 4      | 12.0 | 099 | A     | Moderate | 2      | 2.7  | 126 | A     | Mild     | 0      | 12.6 | 153 | O     | Mild     | 2      | 1.8  |
| 019 | A     | Severe   | 2      | 22.3 | 046 | A     | Moderate | 3      | 19.2 | 073 | B     | Mild     | 3      | 1.1  | 100 | O     | Mild     | 3      | 10.8 | 127 | A     | Mild     | 3      | 5.8  | 154 | O     | Moderate | 2      | 2.6  |
| 020 | O     | Mild     | 1      | 9.3  | 047 | O     | Mild     | 1      | 1.2  | 074 | O     | Moderate | 2      | 18.0 | 101 | A     | Severe   | 0      | 18.3 | 128 | O     | Severe   | 0      | 10.4 | 155 | A     | Moderate | 3      | 12.1 |
| 021 | A     | Moderate | 2      | 15.4 | 048 | A     | Severe   | 1      | 14.2 | 075 | A     | Moderate | 0      | 18.1 | 102 | O     | Mild     | 3      | 1.8  | 129 | B     | Moderate | 0      | 1.6  | 156 | A     | Mild     | 4      | 6.9  |
| 022 | A     | Severe   | 2      | 11.7 | 049 | O     | Moderate | 1      | 12.8 | 076 | AB    | Severe   | 0      | 18.0 | 103 | A     | Severe   | 0      | 2.4  | 130 | O     | Mild     | 4      | 7.9  | 157 | O     | Mild     | 2      | 3.2  |
| 023 | B     | Severe   | 0      | 7.9  | 050 | A     | Mild     | 2      | 12.9 | 077 | B     | Moderate | 2      | 2.0  | 104 | A     | Severe   | 3      | 20.8 | 131 | A     | Severe   | 3      | 3.4  | 158 | A     | Mild     | 0      | 12.2 |
| 024 | A     | Moderate | 3      | 3.2  | 051 | A     | Severe   | 2      | 18.0 | 078 | A     | Mild     | 1      | 8.4  | 105 | AB    | Moderate | 1      | 22.1 | 132 | A     | Mild     | 4      | 18.6 | 159 | O     | Moderate | 2      | 9.3  |
| 025 | O     | Severe   | 3      | 21.3 | 052 | A     | Severe   | 4      | 14.8 | 079 | A     | Mild     | 3      | 8.2  | 106 | B     | Mild     | 4      | 18.9 | 133 | A     | Moderate | 3      | 20.7 | 160 | O     | Moderate | 2      | 13.0 |
| 026 | A     | Moderate | 0      | 13.4 | 053 | O     | Mild     | 1      | 13.8 | 080 | A     | Mild     | 0      | 14.1 | 107 | A     | Moderate | 2      | 11.5 | 134 | B     | Moderate | 4      | 22.6 | 161 | O     | Severe   | 4      | 17.2 |
| 027 | O     | Severe   | 4      | 8.2  | 054 | O     | Moderate | 0      | 4.2  | 081 | O     | Severe   | 4      | 15.5 | 108 | O     | Mild     | 1      | 7.6  | 135 | O     | Mild     | 1      | 17.2 | 162 | A     | Moderate | 3      | 16.6 |

*162* people · one table of observations

# Types of variables

Suppose we collect these data from *162* clinic visitors.

| Person ID  | Blood group | Severity | Visits   | Waiting time (min) |
|------------|-------------|----------|----------|--------------------|
| <i>001</i> | A           | Severe   | <i>4</i> | <i>7.4</i>         |
| <i>002</i> | O           | Mild     | <i>1</i> | <i>2.2</i>         |
| <i>003</i> | B           | Mild     | <i>2</i> | <i>5.4</i>         |
| <i>004</i> | A           | Moderate | <i>2</i> | <i>1.4</i>         |
| <i>005</i> | AB          | Moderate | <i>2</i> | <i>4.6</i>         |
| <i>006</i> | O           | Mild     | <i>3</i> | <i>6.2</i>         |
| <i>007</i> | A           | Mild     | <i>0</i> | <i>3.8</i>         |

# Types of variables

Suppose we collect these data from *162* clinic visitors.

|            | Nominal     | Ordinal  | Discrete | Continuous         |
|------------|-------------|----------|----------|--------------------|
| Person ID  | Blood group | Severity | Visits   | Waiting time (min) |
| <i>001</i> | A           | Severe   | <i>4</i> | <i>7.4</i>         |
| <i>002</i> | O           | Mild     | <i>1</i> | <i>2.2</i>         |
| <i>003</i> | B           | Mild     | <i>2</i> | <i>5.4</i>         |
| <i>004</i> | A           | Moderate | <i>2</i> | <i>1.4</i>         |
| <i>005</i> | AB          | Moderate | <i>2</i> | <i>4.6</i>         |
| <i>006</i> | O           | Mild     | <i>3</i> | <i>6.2</i>         |
| <i>007</i> | A           | Mild     | <i>0</i> | <i>3.8</i>         |

# Types of variables

Suppose we collect these data from *162* clinic visitors.

|            | <b>Nominal</b><br>no particular order | <b>Ordinal</b><br>there is an order | <b>Discrete</b><br>separate values | <b>Continuous</b><br>a value on a scale |
|------------|---------------------------------------|-------------------------------------|------------------------------------|-----------------------------------------|
| Person ID  | Blood group                           | Severity                            | Visits                             | Waiting time (min)                      |
| <i>001</i> | A                                     | Severe                              | <i>4</i>                           | <i>7.4</i>                              |
| <i>002</i> | O                                     | Mild                                | <i>1</i>                           | <i>2.2</i>                              |
| <i>003</i> | B                                     | Mild                                | <i>2</i>                           | <i>5.4</i>                              |
| <i>004</i> | A                                     | Moderate                            | <i>2</i>                           | <i>1.4</i>                              |
| <i>005</i> | AB                                    | Moderate                            | <i>2</i>                           | <i>4.6</i>                              |
| <i>006</i> | O                                     | Mild                                | <i>3</i>                           | <i>6.2</i>                              |
| <i>007</i> | A                                     | Mild                                | <i>0</i>                           | <i>3.8</i>                              |

# Numerical summaries: categorical data

Summarise the blood groups of the same *162* people.

| Blood group | $n$        |
|-------------|------------|
| A           | <i>75</i>  |
| B           | <i>15</i>  |
| AB          | <i>6</i>   |
| O           | <i>66</i>  |
| Total       | <i>162</i> |

## Count

Count the records in each category.

*75* people with blood group A

# Numerical summaries: categorical data

Summarise the blood groups of the same 162 people.

| Blood group | $n$ | %   |
|-------------|-----|-----|
| A           | 75  | 46  |
| B           | 15  | 9   |
| AB          | 6   | 4   |
| O           | 66  | 41  |
| Total       | 162 | 100 |

## Percentage

Express each count as a share of the total.

$$\frac{75}{162} \approx 46\%$$

# Visualising categorical data

The same blood-group counts, shown as bars.

| Blood group | <i>n</i> | %   |
|-------------|----------|-----|
| A           | 75       | 46  |
| B           | 15       | 9   |
| AB          | 6        | 4   |
| O           | 66       | 41  |
| Total       | 162      | 100 |



Bar height represents the number of people.

# Numerical summaries: Measures of centre

Return to the table: select waiting time for rows *001-007*.

| Person ID  | Blood group | Severity | Visits   | Waiting time (min) |
|------------|-------------|----------|----------|--------------------|
| <i>001</i> | A           | Severe   | <i>4</i> | <i>7.4</i>         |
| <i>002</i> | O           | Mild     | <i>1</i> | <i>2.2</i>         |
| <i>003</i> | B           | Mild     | <i>2</i> | <i>5.4</i>         |
| <i>004</i> | A           | Moderate | <i>2</i> | <i>1.4</i>         |
| <i>005</i> | AB          | Moderate | <i>2</i> | <i>4.6</i>         |
| <i>006</i> | O           | Mild     | <i>3</i> | <i>6.2</i>         |
| <i>007</i> | A           | Mild     | <i>0</i> | <i>3.8</i>         |

Use these seven observations to work through the calculation.

# Numerical summaries: Measures of centre

Mean waiting time · minutes

*7.4      2.2      5.4      1.4      4.6      6.2      3.8*

$$\bar{x} = \frac{7.4 + 2.2 + 5.4 + 1.4 + 4.6 + 6.2 + 3.8}{7} \approx 4.43$$

Add the seven waiting times, then divide by seven.

# Numerical summaries: Measures of centre

Mean waiting time · minutes

|       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|
| 7.4   | 2.2   | 5.4   | 1.4   | 4.6   | 6.2   | 3.8   |
| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ | $x_7$ |

$$\bar{x} = \frac{x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7}{7} \approx 4.43$$

Give each observation an index.

# Numerical summaries: Measures of centre

Mean waiting time · minutes

7.4      2.2      5.4      1.4      4.6      6.2      3.8

$$\bar{x} = \frac{x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7}{7} \approx 4.43$$

$$\bar{x} = \frac{1}{7} \sum_{i=1}^7 x_i \approx 4.43$$

In general:  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ .

# Numerical summaries: Measures of centre

Median waiting time · minutes



Observed order

# Numerical summaries: Measures of centre

Median waiting time · minutes



Ordered values

# Numerical summaries: Measures of centre

Median waiting time · minutes



The central observation is *4.6* minutes.

# Numerical summaries: Measures of centre

Median waiting time · minutes



Median = 4.6 min

# Numerical summaries: Measures of centre

Median waiting time · eight observations



Keep the ordered values.

# Numerical summaries: Measures of centre

Median waiting time · eight observations



Add row *008*: *21.0* minutes.

# Numerical summaries: Measures of centre

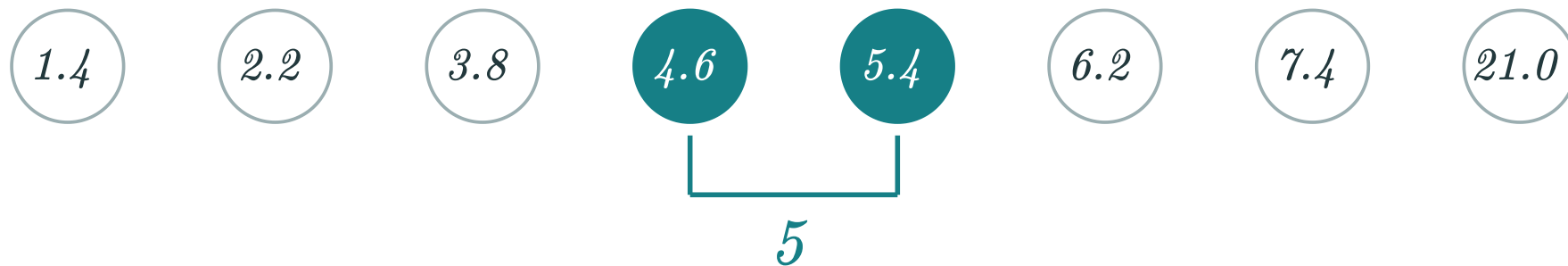
Median waiting time · eight observations



The middle is now between *4.6* and *5.4*.

# Numerical summaries: Measures of centre

Median waiting time · eight observations



$$\text{Median} = \frac{4.6 + 5.4}{2} = 5.0 \text{ min}$$

# Numerical summaries: Measures of centre

Sensitivity to an extreme value



Mean

*6.5*

Median

*5.0*

Mean: *6.5* minutes. Median: *5.0* minutes.

# Numerical summaries: Measures of centre

Sensitivity to an extreme value



Mean

*29*

Median

*5.0*

Only the longest wait changed: *21.0* became *201.0* minutes.

# Numerical summaries: Measures of dispersion

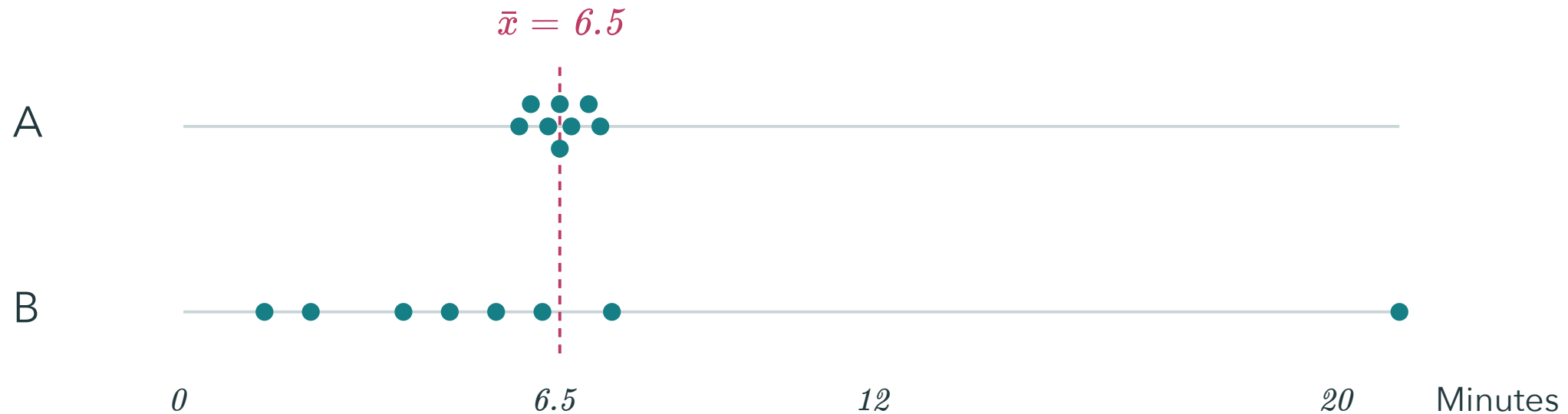
## Standard deviation (SD)

|   |     |     |     |     |     |     |     |     |                 |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----------------|
| A | 5.8 | 6   | 6.3 | 6.5 | 6.5 | 6.7 | 7   | 7.2 | $\bar{x} = 6.5$ |
| B | 1.4 | 2.2 | 3.8 | 4.6 | 5.4 | 6.2 | 7.4 | 21  | $\bar{x} = 6.5$ |

Two datasets, eight values each, the same mean: 6.5 minutes.

# Numerical summaries: Measures of dispersion

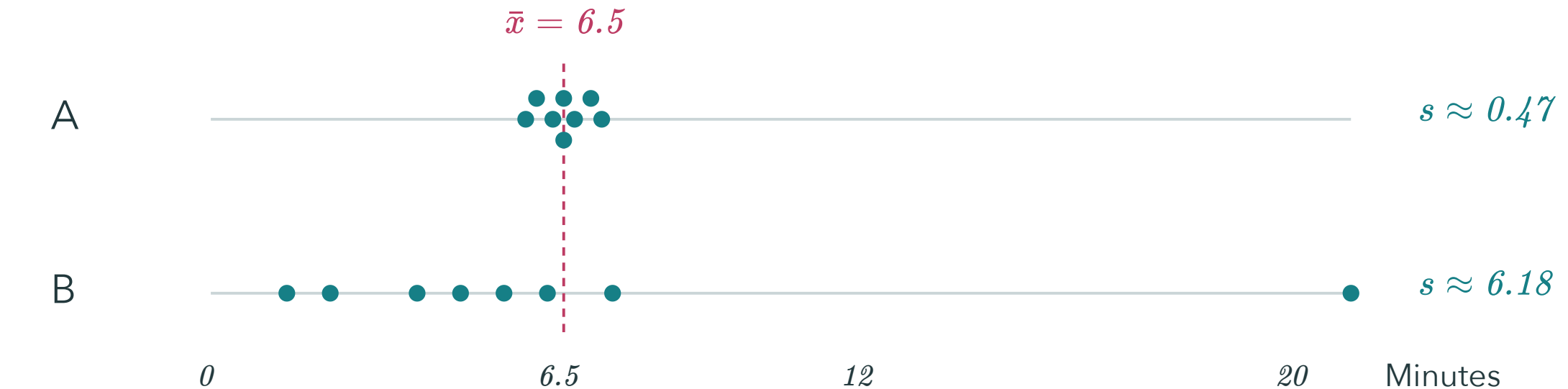
## Standard deviation (SD)



Two datasets, eight values each, the same mean: *6.5* minutes.

# Numerical summaries: Measures of dispersion

## Standard deviation (SD)



$$\text{SD} = s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Larger SD: more spread.  
Identical values:  $s = 0$ .

# Numerical summaries: Measures of dispersion

## Standard deviation (SD)

The article reports both centre and dispersion.

|                                          | All Patients               |                      |
|------------------------------------------|----------------------------|----------------------|
|                                          | Dexamethasone<br>(n = 288) | Placebo<br>(n = 277) |
| Duration of sore throat,<br>mean (SD), d | 3.86 (1.67)                | 3.91 (1.79)          |

Mean

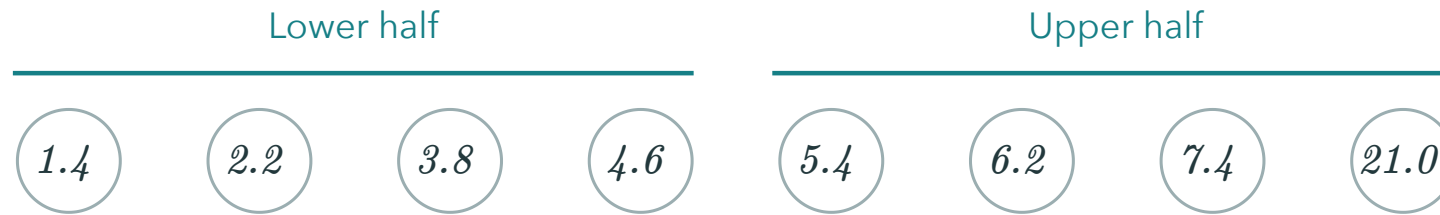
*3.86* days

SD

*1.67* days

# Numerical summaries: Measures of dispersion

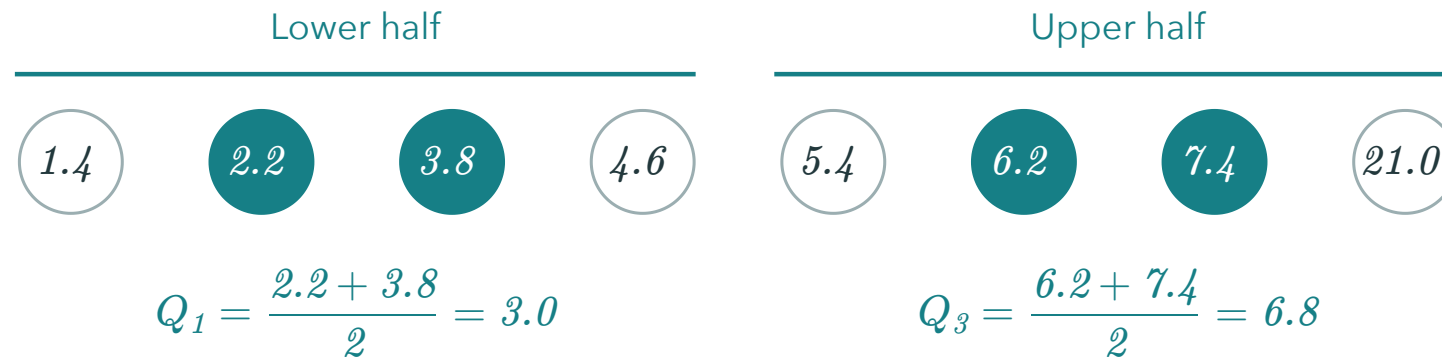
## Quartiles and interquartile range (IQR)



Split the ordered values into two halves.

# Numerical summaries: Measures of dispersion

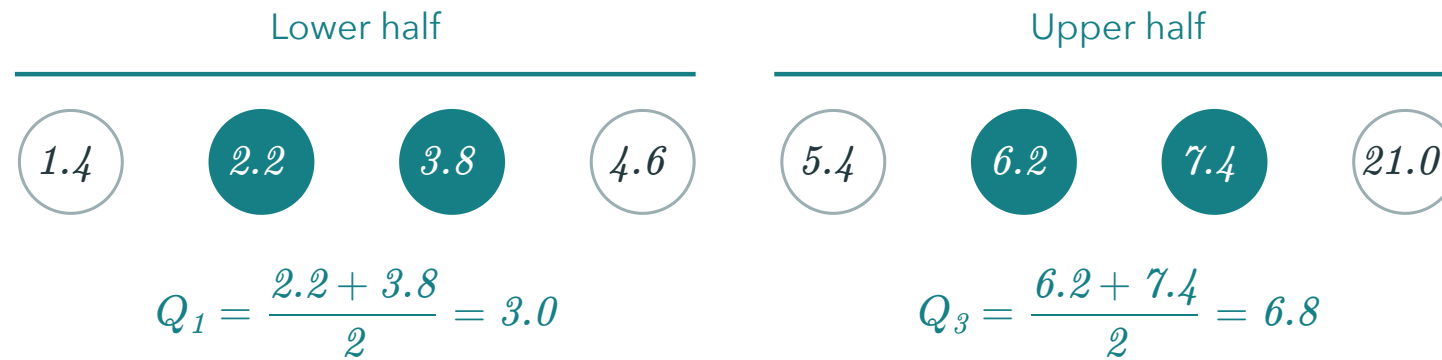
## Quartiles and interquartile range (IQR)



Apply the same median rule in each half.

# Numerical summaries: Measures of dispersion

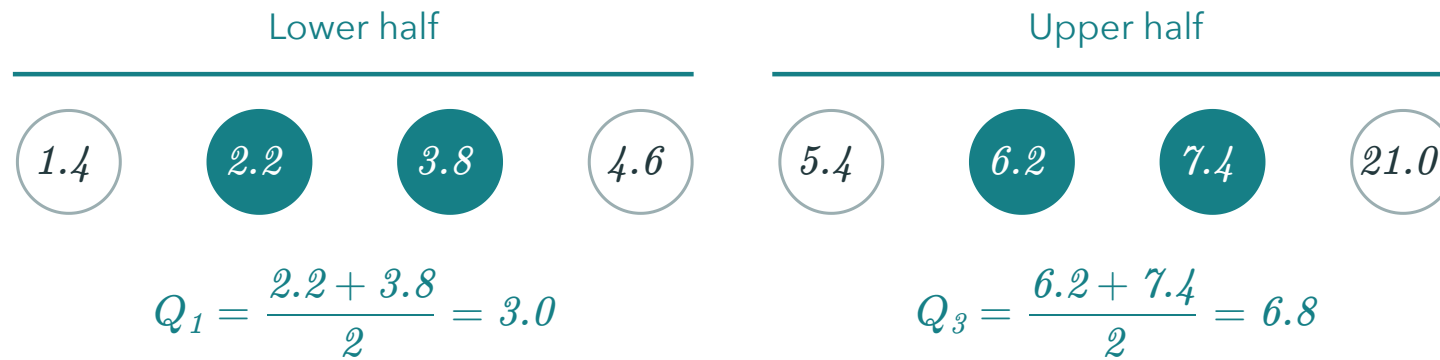
## Quartiles and interquartile range (IQR)



$$\text{IQR} = Q_3 - Q_1 = 6.8 - 3.0 = 3.8 \text{ min}$$

# Numerical summaries: Measures of dispersion

## Quartiles and interquartile range (IQR)



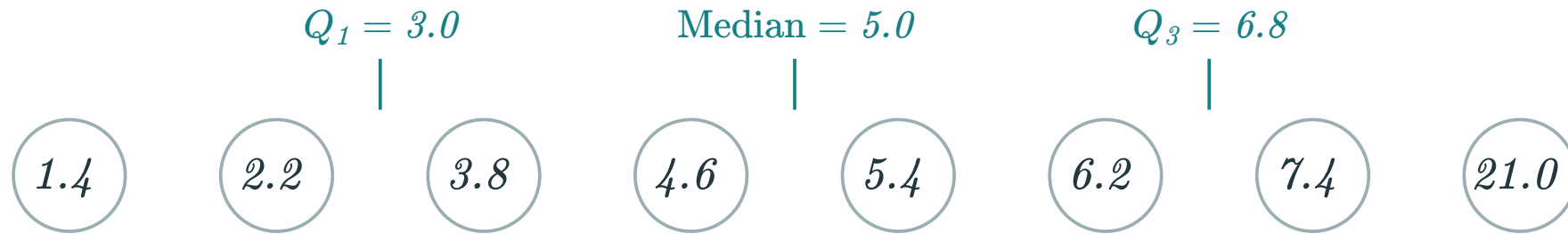
|                      | All Patients               |                      |
|----------------------|----------------------------|----------------------|
|                      | Dexamethasone<br>(n = 288) | Placebo<br>(n = 277) |
| Age, median (IQR), y | 33.7 (26.3-45.8)           | 34.3 (26.0-45.0)     |

Median: 33.7 years

Quartiles: 26.3 and 45.8 years

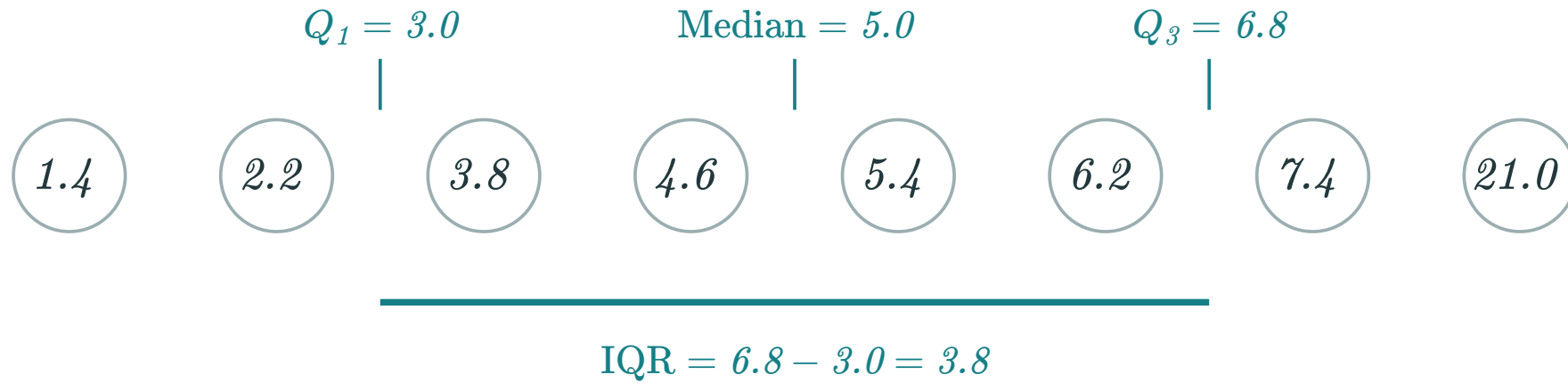
IQR width: 19.5 years

# Visualisation of numerical data: Boxplot



Locate the quartiles and median.

# Visualisation of numerical data: Boxplot



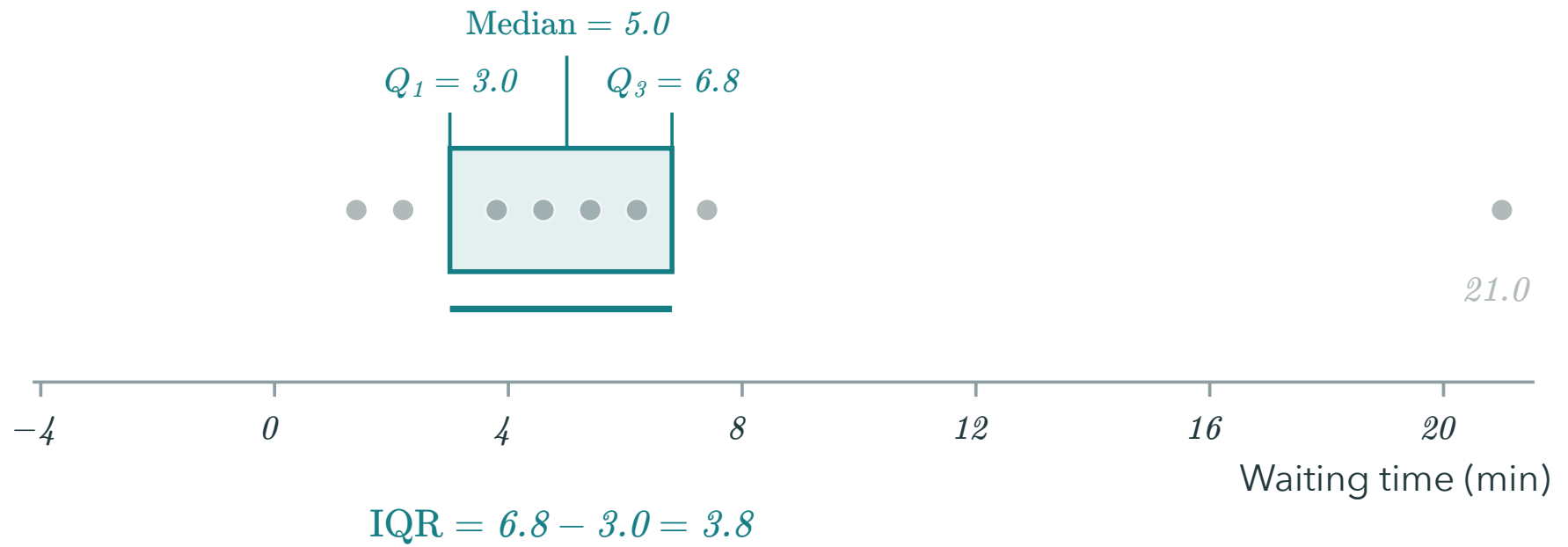
The interquartile range is 3.8 minutes.

# Visualisation of numerical data: Boxplot



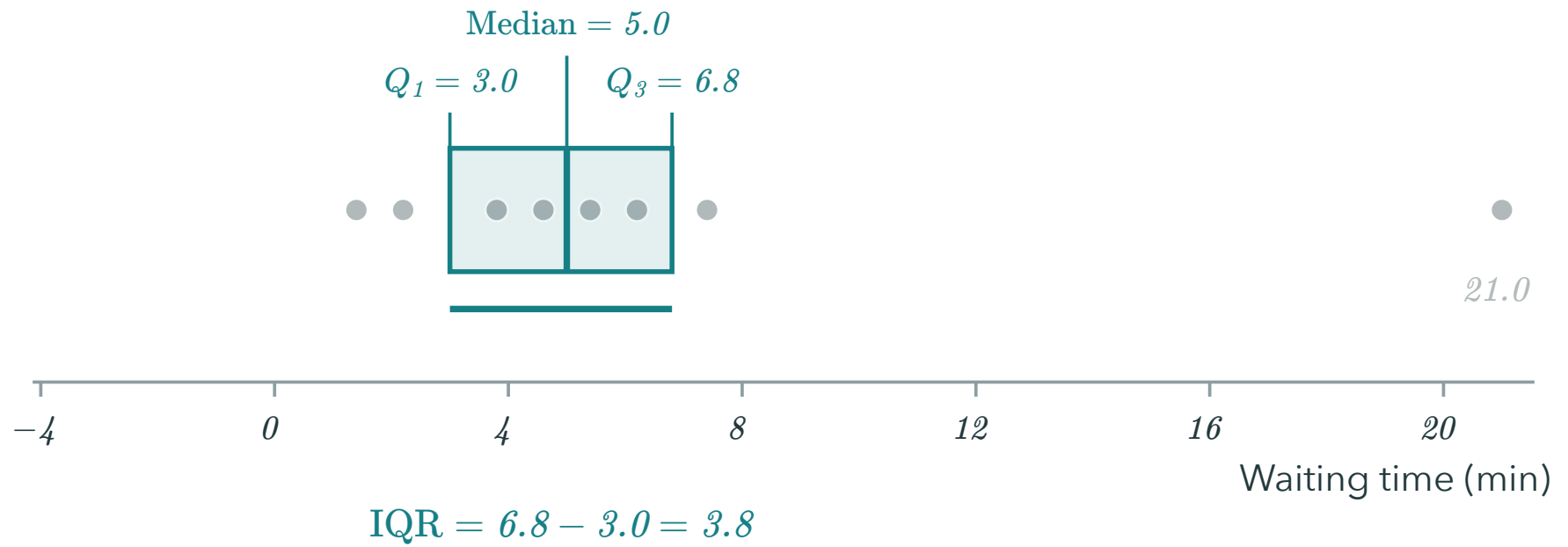
Move the observations onto a numerical scale.

# Visualisation of numerical data: Boxplot



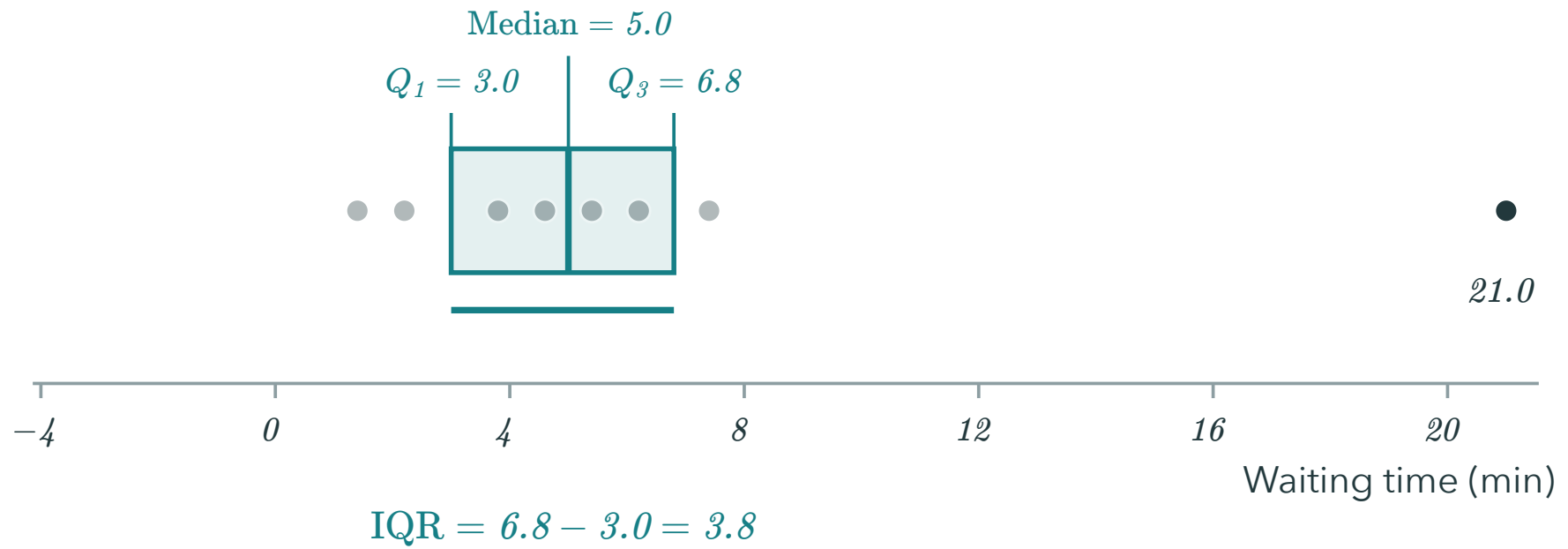
Draw the box from  $Q_1$  to  $Q_3$ .

# Visualisation of numerical data: Boxplot



Add the median.

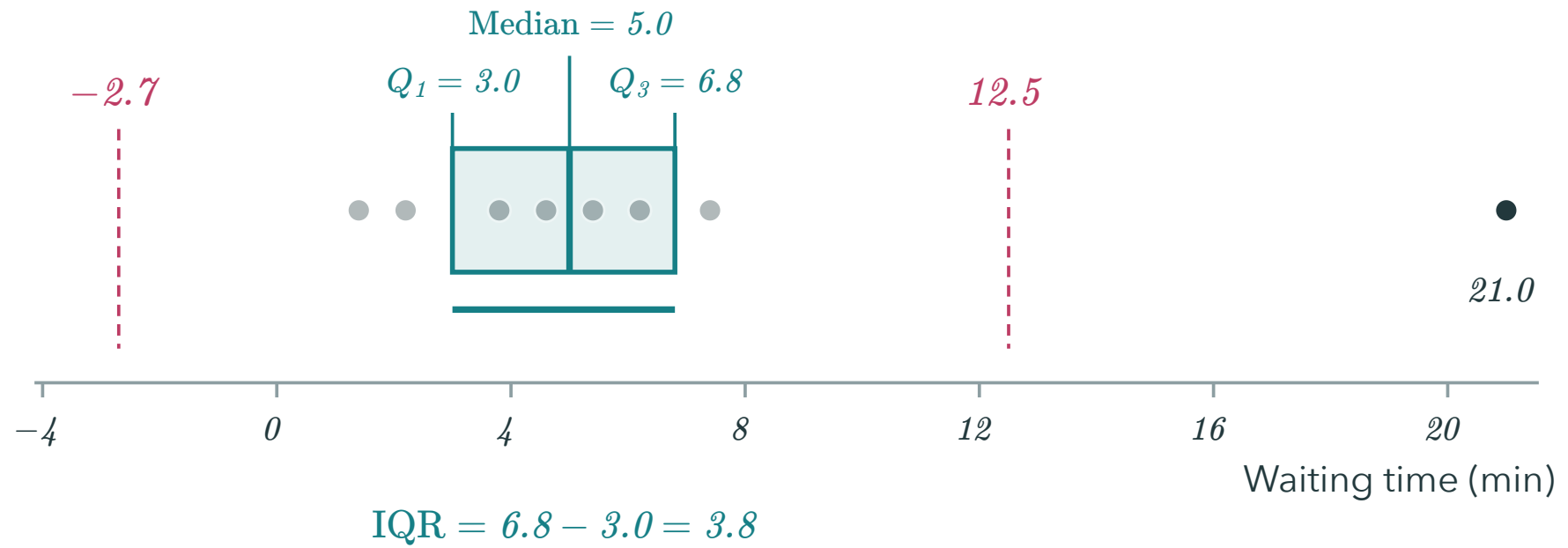
# Visualisation of numerical data: Boxplot



Limits:  $Q_1 - 1.5\text{IQR}$  and  $Q_3 + 1.5\text{IQR}$

The limits are cutoffs, not observed values.

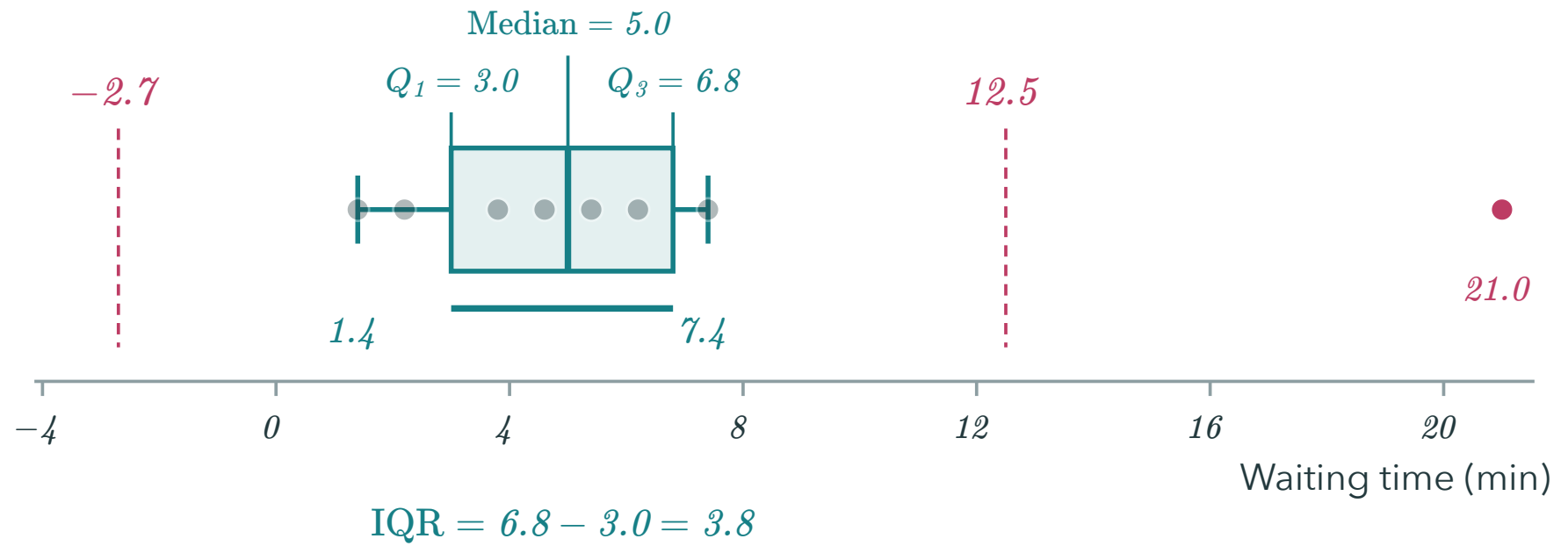
# Visualisation of numerical data: Boxplot



$$3.0 - 1.5 \times 3.8 = -2.7; \quad 6.8 + 1.5 \times 3.8 = 12.5$$

The limits are cutoffs, not observed values.

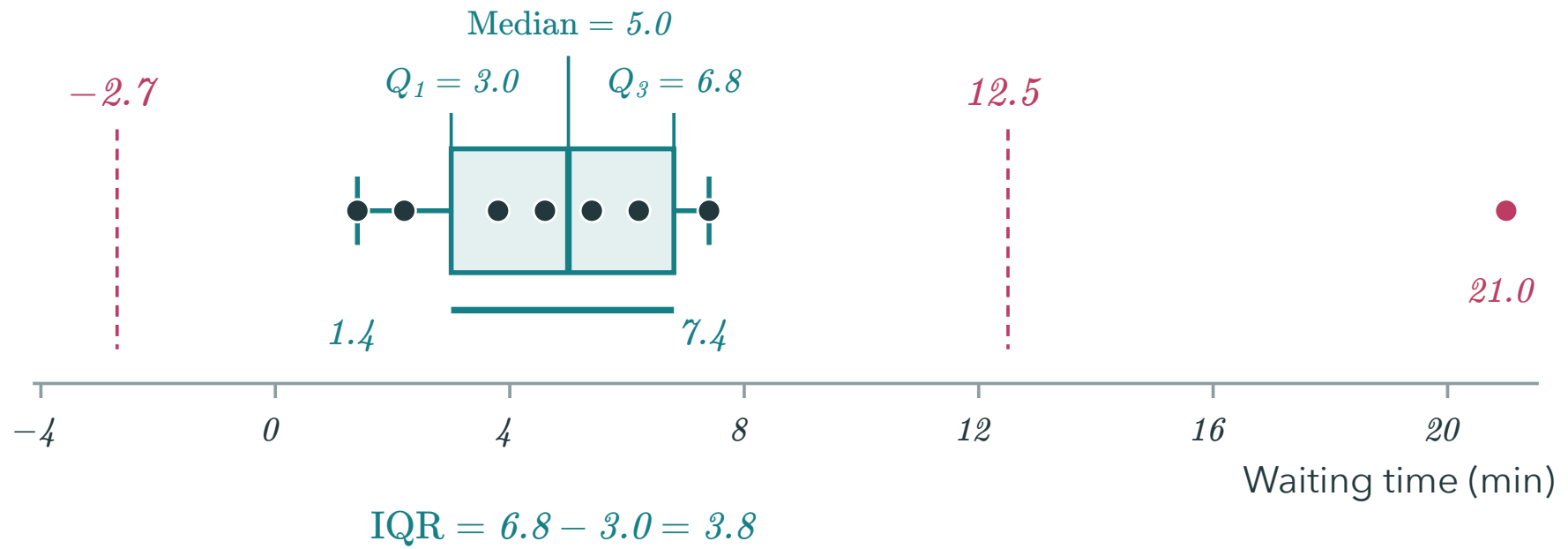
# Visualisation of numerical data: Boxplot



Whiskers reach 1.4 and 7.4 minutes.

21.0 is outside; this does not automatically make it an error.

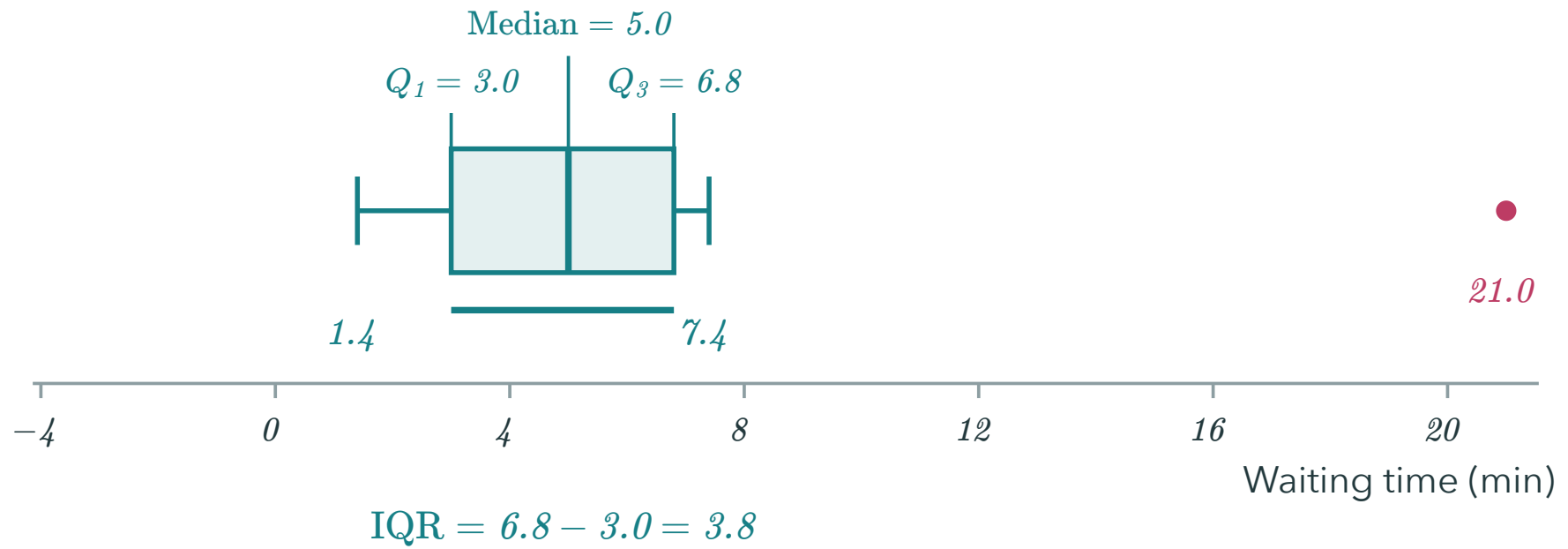
# Visualisation of numerical data: Boxplot



All eight observations over the boxplot.

One point is one observation.

# Visualisation of numerical data: Boxplot



The boxplot retains the point outside the limits.

A boxplot describes the distribution of observations.

# Visualising numerical data: histograms

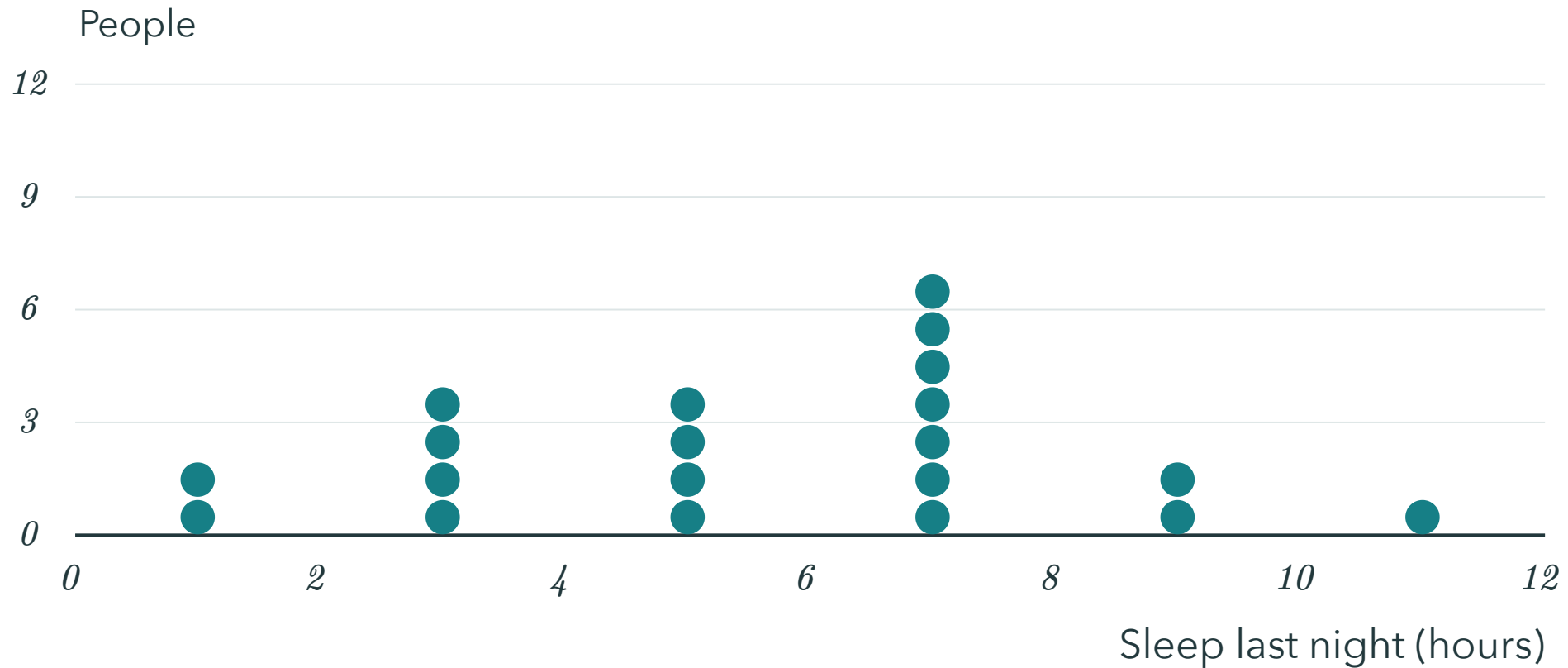
Sleep last night, in hours · 20 people

|        |            |            |            |            |            |            |            |            |            |            |
|--------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Person | <i>1</i>   | <i>2</i>   | <i>3</i>   | <i>4</i>   | <i>5</i>   | <i>6</i>   | <i>7</i>   | <i>8</i>   | <i>9</i>   | <i>10</i>  |
| Hours  | <i>12</i>  | <i>8.5</i> | <i>7.2</i> | <i>7.3</i> | <i>7.7</i> | <i>6</i>   | <i>6.5</i> | <i>4.5</i> | <i>3</i>   | <i>1.2</i> |
| Person | <i>11</i>  | <i>12</i>  | <i>13</i>  | <i>14</i>  | <i>15</i>  | <i>16</i>  | <i>17</i>  | <i>18</i>  | <i>19</i>  | <i>20</i>  |
| Hours  | <i>1.3</i> | <i>2</i>   | <i>2</i>   | <i>3.8</i> | <i>6.6</i> | <i>8.5</i> | <i>5.9</i> | <i>4.6</i> | <i>5.6</i> | <i>6.7</i> |

A table of the recorded measurements.

# Visualising numerical data: histograms

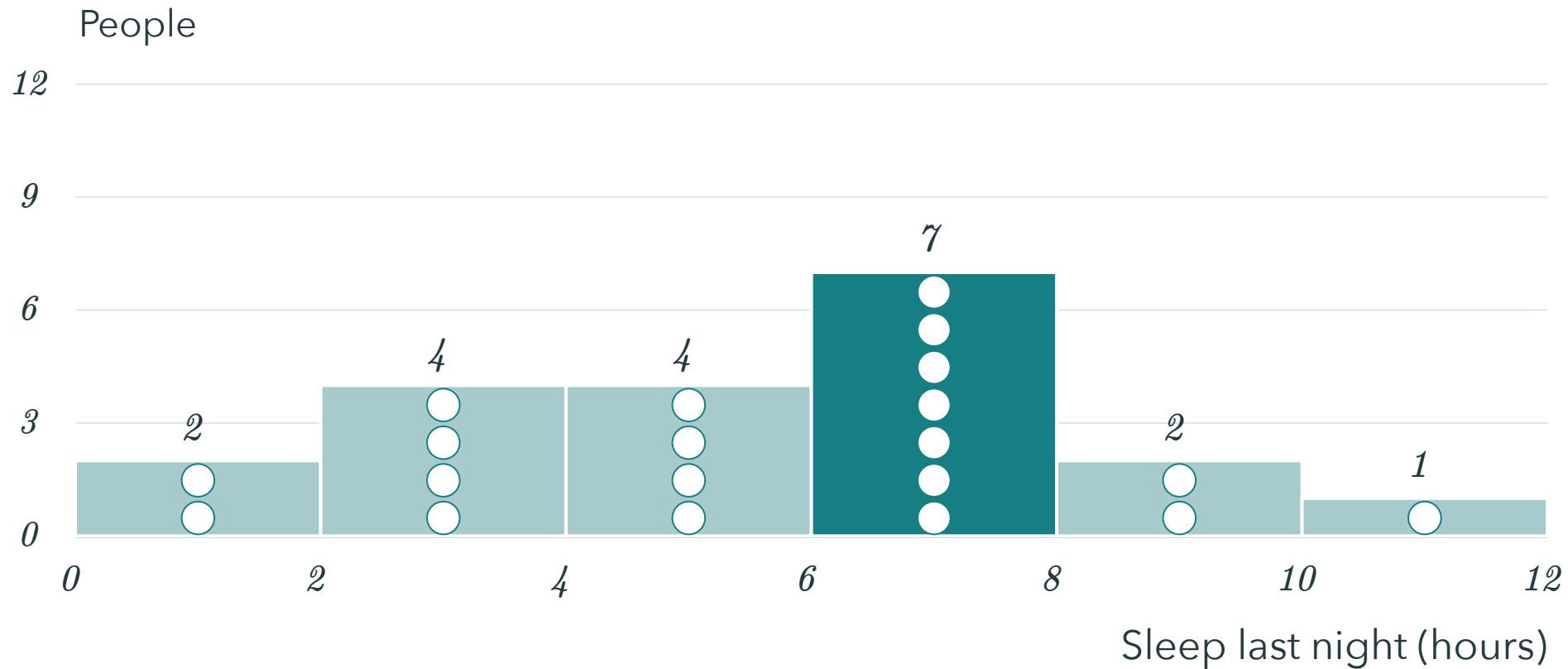
Sleep last night, in hours · 20 people



Group the values into 2-hour intervals.

# Visualising numerical data: histograms

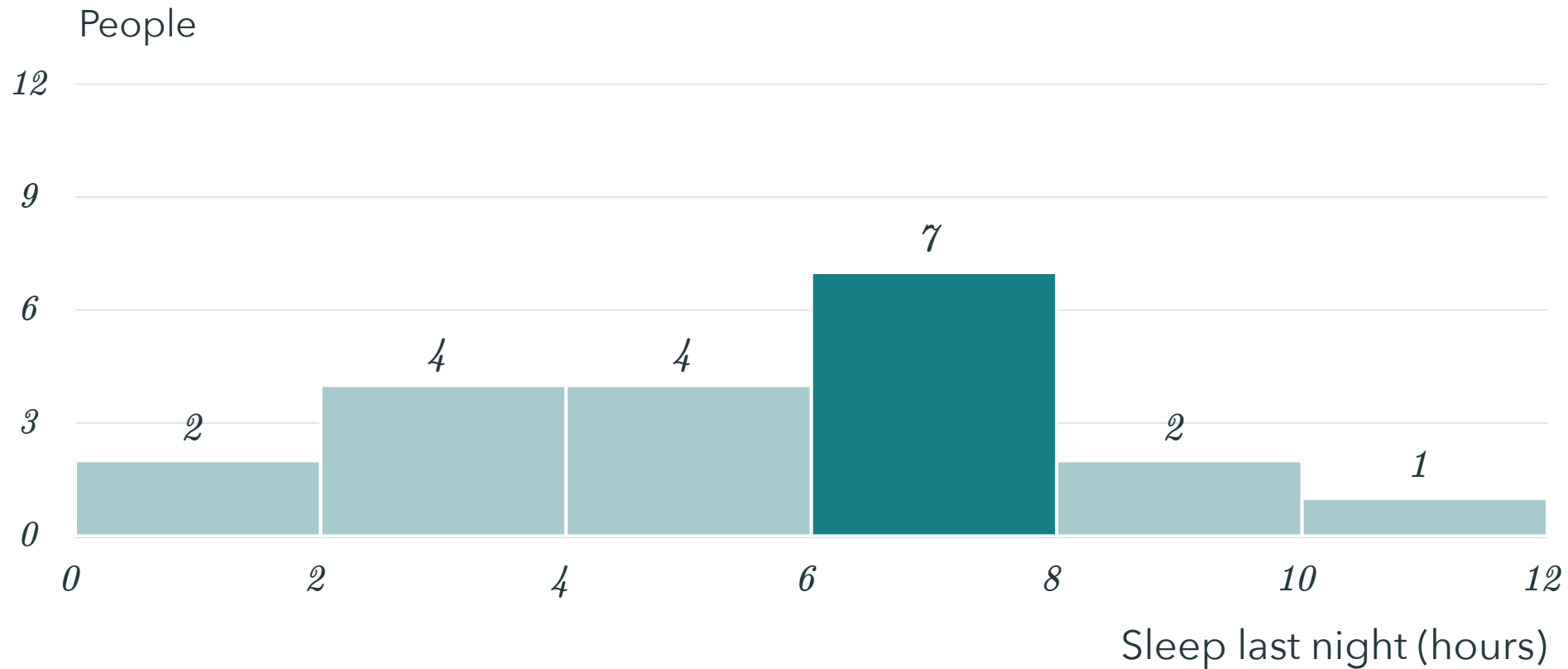
Sleep last night, in hours · 20 people



Count the observations in each interval.

# Visualising numerical data: histograms

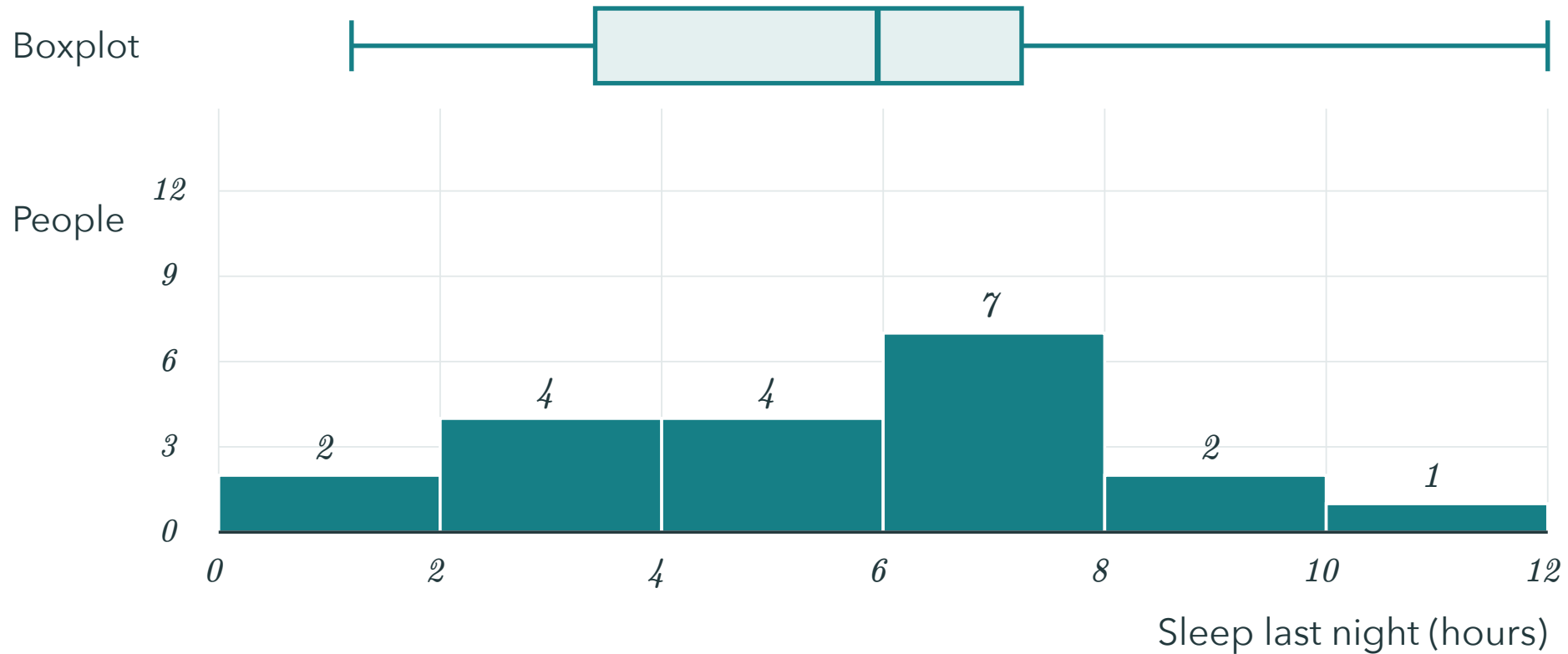
Sleep last night, in hours · 20 people



The bars display those counts.

# Visualising numerical data

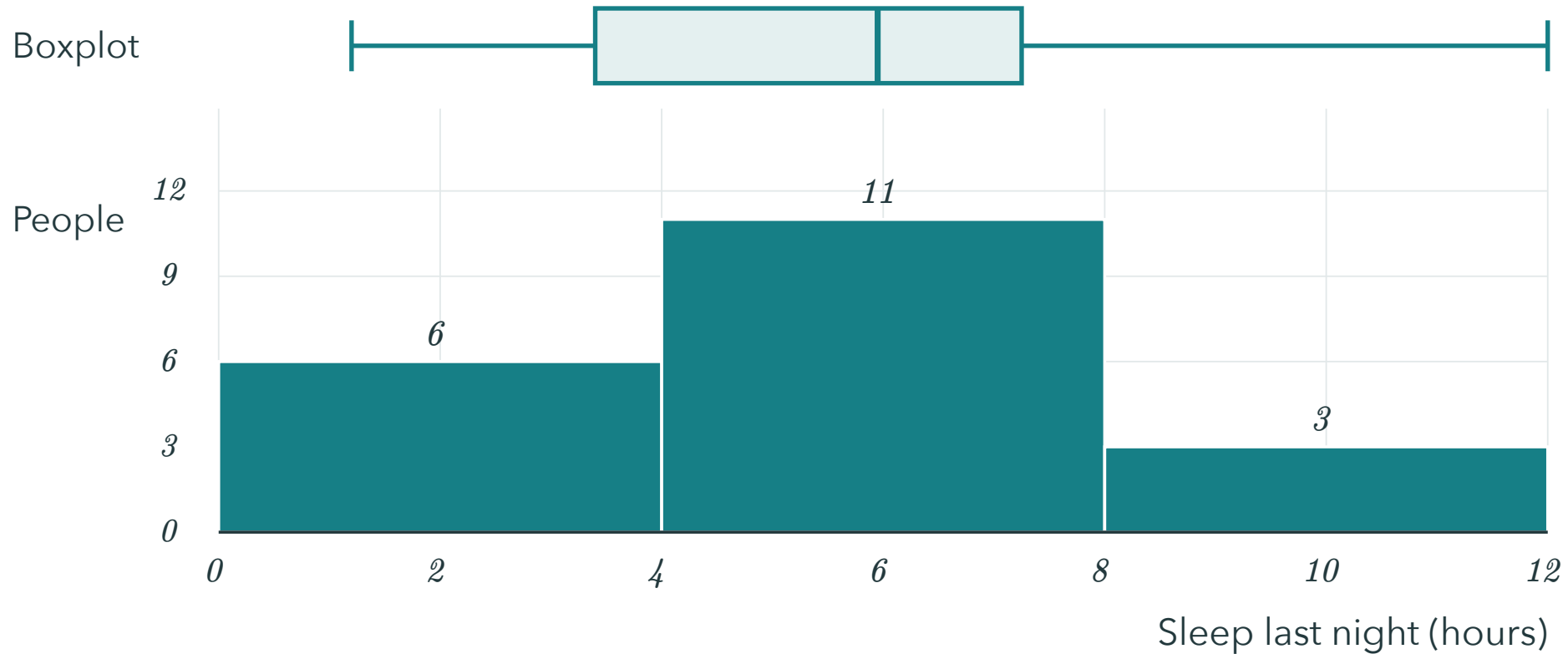
Same 20 sleep measurements · 2-hour intervals



The grouping changes. The boxplot stays the same.

# Visualising numerical data

Same 20 sleep measurements · 4-hour intervals



The grouping changes. The boxplot stays the same.

# Descriptive statistics: interpretation

| Variable                           | Numerical summaries                                       | Visualisation                |
|------------------------------------|-----------------------------------------------------------|------------------------------|
| Categorical<br>nominal / ordinal   | Counts and percentages                                    | Bar charts                   |
| Numerical<br>discrete / continuous | Mean and standard deviation<br>Median and quartiles / IQR | Points, boxplots, histograms |

What do these summaries tell us?

# Descriptive statistics: interpretation

Read the age summary in the placebo group.

|                      | All Patients               |                      |
|----------------------|----------------------------|----------------------|
|                      | Dexamethasone<br>(n = 288) | Placebo<br>(n = 277) |
| Age, median (IQR), y | 33.7 (26.3-45.8)           | 34.3 (26.0-45.0)     |

What do the three numbers describe?

Are *26.0* and *45.0* the youngest and oldest ages?

# Descriptive statistics: interpretation

Read the age summary in the placebo group.

|                      | All Patients               |                      |
|----------------------|----------------------------|----------------------|
|                      | Dexamethasone<br>(n = 288) | Placebo<br>(n = 277) |
| Age, median (IQR), y | 33.7 (26.3-45.8)           | 34.3 (26.0-45.0)     |

Median

*34.3 years*

Quartiles · the middle half

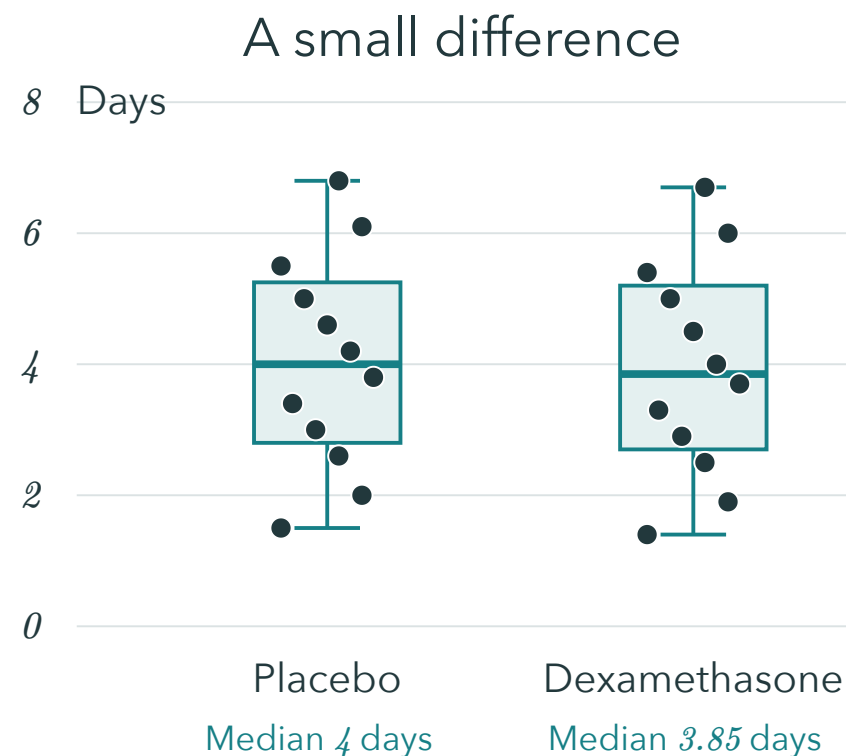
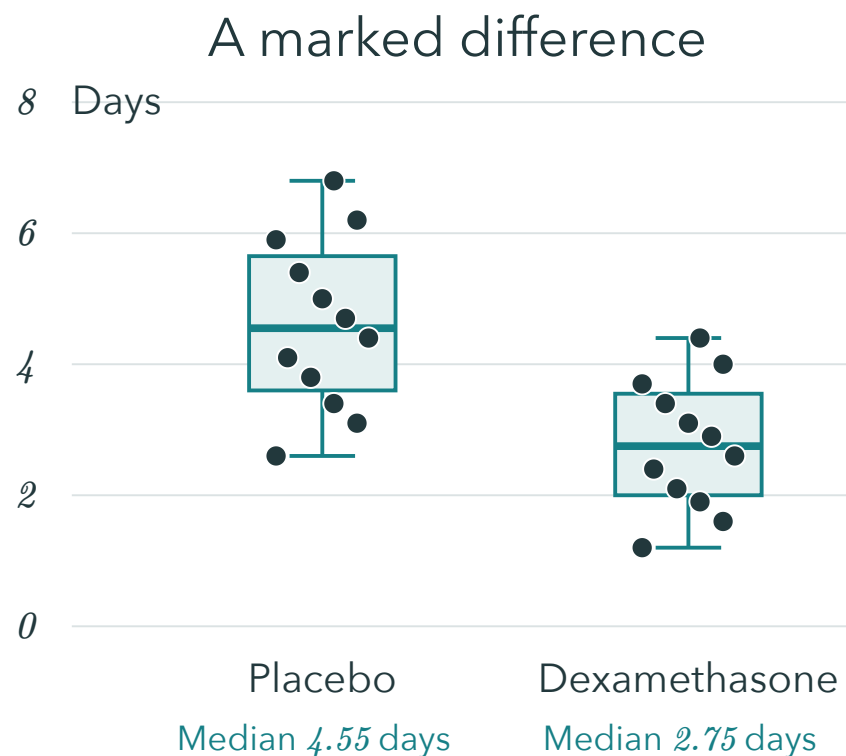
*26.0–45.0 years*

$\text{IQR} = 45.0 - 26.0 = 19.0 \text{ years}$

# Comparing numerical data

Duration of sore throat · days

**Fictional data** · placebo and dexamethasone · two possible observed samples



A plot describes the observed groups. It does not establish statistical significance.

# From description to inference

**CONCLUSIONS AND RELEVANCE** Among adults presenting to primary care with acute sore throat, a single dose of oral dexamethasone compared with placebo did not increase the proportion of patients with resolution of symptoms at 24 hours. However, there was a significant difference at 48 hours.

**This visual difference is descriptive.**

It does not establish statistical significance.

Next: confidence intervals quantify uncertainty about an estimated difference.

# From description to inference

**CONCLUSIONS AND RELEVANCE** Among adults presenting to primary care with acute sore throat, a single dose of oral dexamethasone compared with placebo did not increase the proportion of patients with resolution of symptoms at 24 hours. However, there was a significant difference at 48 hours.

**Inference accounts for sampling variability.**

Next: confidence intervals quantify uncertainty about an estimated difference.

# References

1. Hayward G, Hay AD, Moore MV, et al. Effect of oral dexamethasone without immediate antibiotics vs placebo on acute sore throat in adults: a randomized clinical trial. *JAMA* 2017;317(15):1535. doi:[10.1001/jama.2017.3417](https://doi.org/10.1001/jama.2017.3417)

Thank you very much for your attention